
AI Agents for *Dynamic Healthcare*: World Models, Reasoning, Simulation, and Evaluation in Human-Governed Systems

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Abstract

AI systems for healthcare are moving from static prediction and single-turn question answering toward agents that reason over evolving patient states, use clinical tools, interact with EHR/FHIR, coordinate with clinicians, and recommend bounded actions. This workshop builds a NeurIPS community around clinical agents for *dynamic healthcare*: world models of patient trajectories and care processes, reasoning and planning under uncertainty, simulation environments for training and stress testing, dynamic evaluation beyond static QA, and human-governed deployment with auditability, oversight, privacy, safety, and workflow integration.

1 Main Proposal

Introduction. Recent advances in large language models, multimodal foundation models, and agentic AI systems have shown generalizable capabilities across medical reasoning, visual recognition, and interactive decision support [1–8]. These developments are especially consequential for healthcare, where clinical work is dynamic, multimodal, longitudinal, and action-oriented: clinicians integrate evolving patient histories, imaging findings, and treatment trajectories under uncertainty, time pressure, and competing priorities [1, 3, 4, 9, 10]. AI agents thus hold real promise for augmenting clinical workflows such as diagnosis support, care planning, patient communication, and care coordination [11, 12, 5, 6, 13, 14]. Yet realizing this potential requires moving beyond static medical question answering toward *clinical world models, dynamic evaluation protocols, and human-governed systems that ensure safety, auditability, and responsible workflow integration* [13–16, 10, 17].

Considering healthcare is fundamentally dynamic, this workshop focuses on **AI agents for dynamic healthcare**: agents that reason over evolving clinical context and recommend bounded, auditable actions from LLM-based assistants to multi-agent clinical teams. Critically, the workshop studies *how agents can safely assist clinicians, patients, and hospital systems with human governance*, questions prerequisite to safe deployment, yet largely unaddressed by current evaluation paradigms.

Objectives. AI agents for dynamic healthcare represent a frontier where machine learning, clinical medicine, and systems safety converge. This workshop establishes a focused forum to advance agentic systems that reason over evolving patient states, interact with clinical tools, and remain governed by human oversight, moving the field beyond static prediction toward world models, simulation, and dynamic evaluation. Specifically, the workshop has the following objectives:

- **Advance clinical AI agents for dynamic healthcare.** We encourage research on agents that integrate longitudinal patient data, clinical tools, and workflow constraints to support diagnosis, care planning, documentation, and coordination.

- **Connect agentic AI with clinical world models.** We bring together researchers on patient trajectory modeling, disease simulators, and digital twins to explore how such environments support training, testing, and interpretation of clinical agents.
- **Deepen understanding of reasoning and planning in longitudinal settings.** We explore how agents should update beliefs under uncertainty, use tools safely, recover from errors, and defer to clinicians in dynamic and partially observed clinical environments.
- **Advance evaluation beyond static medical QA.** We highlight benchmarks and protocols that measure longitudinal consistency, action appropriateness, and failure recovery, prioritizing clinically meaningful evaluation over aggregate performance.
- **Strengthen safety, governance, and human oversight.** We discuss bounded autonomy, clinician override, audit trails, and regulatory readiness, emphasizing human-governed systems over fully autonomous decision makers in high-stakes clinical settings.
- **Foster cross-disciplinary collaboration.** Through talks, panels, and breakout discussions, we aim to unite ML researchers, clinicians, and health informaticians around shared benchmarks, open environments, and reporting standards.

Motivation. LLM-based agent systems increasingly demonstrate tool use, multimodal reasoning, and EHR-style workflows in realistic settings. But evaluation is lagging: static medical QA does not capture longitudinal consistency, action appropriateness, or failure recovery, let alone alignment with clinical utility and regulatory expectations. Without a coordinating research community, the field risks fragmenting into incomparable simulators and benchmarks that reward fluent answers over safe clinical behavior. NeurIPS is the right venue because this problem sits at the intersection of agents, reinforcement learning, world models, generative modeling, uncertainty quantification, causal inference, and human-AI collaboration, which is a community uniquely positioned to close the gap between impressive model capabilities and trustworthy clinical deployment.

Encouraged topics. This workshop offers an interdisciplinary platform for researchers and practitioners to present their latest work at the intersection of agentic AI, clinical medicine, and healthcare systems. Key topics of interest include, but are not limited to, the following:

- **Clinical AI agents for dynamic healthcare.** Agentic systems for diagnosis, triage, documentation, monitoring, care planning, and coordination, including LLM agents, multimodal agents, EHR/FHIR agents, and multi-agent clinical teams.
- **Medical reasoning and planning in longitudinal care.** How agents reason over time, update beliefs under uncertainty, use evidence, and defer safely to clinicians as states evolve.
- **Clinical world models and patient simulation.** Patient trajectory models, disease simulators, digital twins, and uncertainty-aware environments that serve as testbeds for training, evaluating, stress-testing, and interpreting clinical agents before deployment.
- **EHR/FHIR environments and clinical tool use.** Realistic environments where agents interact with health records, retrieval systems, and ordering tools, with emphasis on safe tool use, audit logging, and the gap between benchmarks and real hospital systems.
- **Dynamic evaluation beyond static medical QA.** Benchmarks and protocols measuring longitudinal consistency, action appropriateness, calibration, failure recovery, and clinically meaningful failure modes, beyond aggregate accuracy.
- **Safety, reliability, and bounded autonomy.** Auditable, privacy-preserving, and controllable agents. Topics include clinician override, hallucination mitigation, adversarial robustness, deployment monitoring, regulatory readiness, and post-deployment incident response.
- **Human-agent collaboration and workflow integration.** Clinician-in-the-loop design, care-team coordination, trust, workflow burden, escalation, and prospective or retrospective deployment studies in real clinical settings with measurable clinical outcomes.
- **Datasets, tools, and reproducibility.** Benchmark environments, synthetic patient datasets, annotation protocols, reporting standards, and open-source simulators that enable fair comparison under shared safety assumptions and transparent experimental documentation.
- **Governance, regulation, and position papers.** Critical discussion on regulatory pathways, accountability, health equity, and responsible deployment. Position papers are welcome when they identify unsolved risks or propose evaluation standards.

Workshop format and logistics. We propose a one-day in-person workshop with 7–9 hours of content. The program will include invited talks, contributed talks, posters, demos, a structured debate, a panel, and breakout discussions. For a tentative structure and the scheduled submission process and timeline, please consult our website.

Attendance. We expect 200–250 in-person attendees, 200–400 online attendees, 100–150 submissions, 30–40 posters, 6–8 contributed talks, and 6–10 demos or benchmark presentations. The audience will include researchers in agents, RL, world models, foundation models, medical AI, clinical NLP, imaging, informatics, HCI, privacy, safety and responsible deployment, as well as clinicians and health-system collaborators.

Tentative list of invited speakers. All invited speakers have accepted and confirmed in-person availability: **Prof. James Zou** (Stanford University), **Prof. Xiangliang Zhang** (University of Notre Dame), **Prof. Alan Yuille** (Johns Hopkins University), **Prof. Mahnaz Maddah** (Broad Institute of MIT and Harvard), **Prof. Jonathan H. Chen** (Stanford Medicine), and **Daguang Xu** (NVIDIA Research). Together, they span biomedical data science, trustworthy AI, AI for science, agent systems, computer vision, medical imaging, EHR-based learning, and healthcare AI infrastructure.

Diversity Statement. Diversity, equity, inclusion, and accessibility are central to this workshop, as safe AI agents for dynamic healthcare require insights from multiple technical, clinical, and institutional perspectives. We promote inclusivity by assembling a team and speaker lineup spanning varied disciplines, backgrounds, career stages, and methodologies. More details are listed as follows.

Organizers. We have assembled a team with complementary expertise across institutions, regions, career stages, and disciplines. **(a) Geographic and institutional breadth.** The organizers bring perspectives from North America and Asia and from medical schools, hospitals, engineering programs, and international AI institutes, reducing reliance on any single institutional or regional viewpoint. **(b) Inclusive representation.** The team includes researchers across different backgrounds and career paths, with balanced participation across genders and seniority levels. **(c) Academic seniority.** The committee spans postdoctoral researchers, assistant professors, and senior faculty, combining emerging organizers with established leaders. **(d) Disciplinary breadth.** Our team bridges core machine learning, biomedical engineering, health data science, and clinical informatics. **(e) Research specialization.** Organizers cover agent systems, multimodal foundation models, EHR-based learning, medical imaging, clinical decision support, generative modeling, world models, and AI safety/privacy.

Keynote Speakers. We have curated a diverse invited-speaker lineup representing the AI-healthcare ecosystem. **(a) Affiliation Variety.** Confirmed speakers represent distinct academic centers and traditions. **(b) Thematic Scope.** Invitees span trustworthy AI, biomedical data science, computer vision, medical image analysis, and clinical decision support. **(c) Translational Perspectives.** The program balances algorithmic researchers with clinical translation leaders, connecting technical advances in agents with real-world healthcare constraints.

Inclusive Participation Mechanisms. Beyond personnel, we foster equity through workshop design. We use open reviewer self-nominations and varied participation avenues, including contributed talks, posters, demos, lightning talks, and mentoring-oriented reviews, designed to empower junior researchers, clinicians, and contributors from underrepresented institutions.

Location preference, sponsor and technical needs, and earlier versions. Our location preference is **Atlanta**. We are in discussion with NVIDIA and GE Healthcare about possible sponsorship to support community building and awards, including Best Paper and Outstanding Reviewer Awards.

Earlier versions. As an emerging frontier, no prior workshops have specifically addressed the topic of AI agents for dynamic healthcare, emphasizing the novelty and growing interest in this interdisciplinary field. As a pioneering area, it holds enormous potential in revolutionizing medical practices, particularly with the rapid progress in agentic systems, world models, and longitudinal reasoning. While previous venues have explored static medical foundation models, this inaugural edition uniquely makes dynamic clinical agency the central object of study. Furthermore, it clearly distinguishes itself from general world-model or agent workshops by strictly grounding world models, reasoning, simulation, and evaluation in healthcare workflows, clinical oversight, and patient safety.

Other comments. This workshop will examine how agentic systems and clinical world models can support dynamic healthcare. Our goal is to foster reliable, collaborative medical agents that reason over evolving patient trajectories and recommend bounded actions. By bringing together researchers and clinicians, we aim to create a shared vocabulary and roadmap beyond static medical answers toward practical clinical assistance. We emphasize the critical aspects of auditability, uncertainty awareness, and human governance, as they are crucial for gaining clinician trust, ensuring patient safety, and facilitating the responsible deployment of AI in healthcare workflows.

2 Organization Information

Organizer list.

- **Yichen Wu** (MGB/Harvard Medical School). Email: yiwu6@mgh.harvard.edu. Wu is a postdoctoral research fellow working on continual learning, multimodal large language models, and clinical agents. He will coordinate the website and CFP, manage submission logistics, and lead the clinical agent and dynamic evaluation tracks.
- **Yuhan Liu** (Mohamed bin Zayed University of Artificial Intelligence). Email: yuhan.liu@mbzuai.ac.ae. Liu is a postdoctoral research fellow working on trustworthy agentic AI, agent simulation and multimodal reasoning. He will coordinate OpenReview, reviewer assignment, mentoring activities and lead the agent safety and Human-agent collaboration and workflow integration tracks.
- **Xiang Li** (Massachusetts General Hospital and Harvard Medical School). Email: xli60@mgh.harvard.edu. Li is an assistant professor working on medical foundation models, generative AI, expert-machine alignment, medical informatics, and counterfactual causal inference. He will oversee clinical relevance and reviewer quality, coordinate invited speakers and advisory input, and lead the Medical reasoning and Dataset tracks.
- **Wengong Jin** (Northeastern University and Broad Institute of MIT and Harvard). Email: w.jin@northeastern.edu. Jin is an assistant professor working on geometric and generative AI models for drug discovery, biology, and molecular design. He will lead the world models and governance tracks, with emphasis on simulation-based evaluation.
- **Ruogu Fang** (Vanderbilt University). Email: ruogu.fang@vanderbilt.edu. Fang works on AI-empowered precision brain health, biomedical imaging, health data science, and brain/bio-inspired trustworthy AI. She will coordinate diversity and mentoring, clinical evaluation, and reviewer recruitment from biomedical engineering and health AI.
- **Marzyeh Ghassemi** (MIT EECS and IMES). Email: mghassemi@mit.edu. Ghassemi works on robust, private, and fair machine learning for health and has extensive service and organizing experience in health ML. She will provide senior leadership on safety, fairness, governance, COI policy, reviewer standards, and community outreach.

Advisory Board. The advisory board will provide senior guidance on clinical relevance, technical scope, and community outreach. Prof. **Quanzheng Li** (Massachusetts General Hospital and Harvard Medical School) will advise on medical imaging, clinical data infrastructure, hospital workflow constraints, and evaluation grounded in real clinical settings. Prof. **Jimeng Sun** (University of Illinois Urbana-Champaign and Carle Illinois College of Medicine) will advise on AI for healthcare, longitudinal clinical modeling, treatment recommendation, and reviewer recruitment from the health AI community. Prof. **Regina Barzilay** (MIT CSAIL and Jameel Clinic) will advise on trustworthy clinical AI, drug discovery, translational machine learning, and connections to the broader biomedical AI community. Together, the advisors will help refine the CFP, identify invited speakers and reviewers, strengthen clinical relevance, safety, and evaluation standards, and support community building across ML, medicine, and health systems.

Related Experience The organizers have extensive experience hosting related events and serving ML and biomedical AI communities, including:

- Advancements in Medical Foundation Models (AIM-FM) **WORKSHOP** at **NeurIPS 2024**.
- Multiscale Multimodal Medical Imaging (MMMI) **WORKSHOP SERIES** at **MICCAI 2019, 2022–2024**.
- SAGES Critical View of Safety **CHALLENGE** at **MICCAI 2024**.
- Machine Learning in Drug Discovery **SYMPOSIUM COMMITTEE/CO-CHAIR** at the **Broad Institute 2022–2025**.
- Women in MICCAI (WiM) **PRESIDENTIAL SERVICE** within the **MICCAI Society 2023–2025**.
- Program Committee and Area Chair **SERVICE** at **MICCAI 2021–2024**.
- Sparsity Techniques in Medical Imaging **WORKSHOP** at **MICCAI 2014**.

- Machine Learning for Health (ML4H) **WORKSHOP** at **NIPS/NeurIPS 2016–2018**.
- Conference on Health, Inference, and Learning (CHIL) **GENERAL CHAIR** at **ACM CHIL 2020–2021**.
- Women in Machine Learning (WiML) **WORKSHOP ORGANIZATION** at **NIPS 2014**.
- Reviewing, area-chair/session-chair, and **PC SERVICE** at **NeurIPS, ICLR, ICML, ACL, EMNLP, IJCAI, AAAI, CVPR, and ICCV**.

Together, this experience covers workshop design, reviewer recruitment, submission management, invited-speaker coordination, mentoring, diversity initiatives, and community building.

Conflict-of-interest management. We will follow NeurIPS workshop COI requirements. Organizers will not submit papers to this workshop, and people with personal COIs with organizers, including their PhD students and postdocs, will not submit papers. Organizers will not assess submissions from their own institution, collaborators, or personal conflicts. Reviewer assignments will be checked for institutional, advisor/advisee, collaborator, and personal conflicts. Invited speakers may participate but will not review conflicted submissions. We are discussing possible sponsorship with NVIDIA and GE HealthCare; any sponsorship, if finalized, will be firewalled from scientific review, reviewer assignment, acceptance decisions, awards, confidential submissions, reviews, reviewer identities, internal discussions, and award deliberations.

Program committee. We expect 100–150 submissions and will recruit approximately 120–160 reviewers, plus a small reserve pool, so that each submission receives at least three reviews and reviewer load remains limited. Reviewers will be matched to submissions using topic expertise, author conflicts, reviewer bids when available, and manual organizer checks. We will ask reviewers to assess technical quality, clinical relevance, safety implications, and fit to the workshop theme.

Preliminary PC/reviewer pool will emphasize postdoctoral researchers, senior PhD students, research scientists, and junior faculty with hands-on expertise in clinical agents, medical AI, EHR/FHIR systems, medical imaging, world models, safety, and human-AI collaboration. Senior researchers from the organizers’ and advisory board’s networks will be invited to serve as senior PC members, meta-reviewers, or reviewer-recruitment advisors where appropriate. Initial senior PC/reviewer candidates include Samuel Schmidgall, Michael Moor, Irene Chen, Danielle Bitterman, Shalmali Joshi, Riccardo Miotto, Xiaoxiao Li, Shadi Albarqouni, Ismini Lourentzou, Linwei Wang, Zhen Chen, Xiaoqing Guo, Xiaohan Xing, Wuyang Li, Yu Jiang, Chen Yang, Pengyu Wang, Zhihao Peng, Bo Zhao, Zhiyi Shi, Jinqian Yang, and Minjun Kim. We will expand this pool through qualified trainees, former students, collaborators, and related research groups, while enforcing COI checks for institutional, advisor/advisee, collaborator, and personal conflicts.

Reviewer recruitment will prioritize expertise in clinical agents, clinical NLP, EHR/FHIR systems, medical imaging, multimodal learning, reinforcement learning, world models, simulation, human-AI collaboration, health AI safety, privacy, and governance. We will expand the PC through organizer and advisory-board networks, including qualified collaborators, former students, trainees, and related research groups, while enforcing COI checks for institutional, advisor/advisee, collaborator, and personal conflicts. We will also recruit from prior health AI workshop communities, NeurIPS/ICLR/ICML health AI authors, CHIL/ML4H communities, open reviewer self-nominations, and relevant clinical informatics networks. Senior organizers and advisory-board members will help audit reviewer coverage, resolve borderline cases, calibrate review standards, and ensure accepted papers collectively reflect both technical rigor and clinical relevance.

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